

# DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

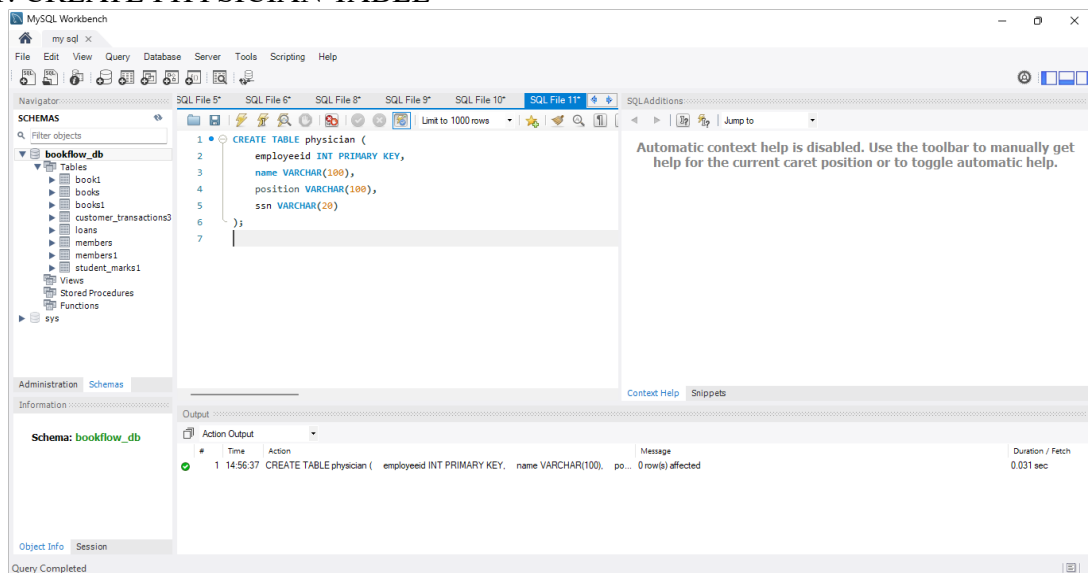
## Sec:- 3

Name:- Srithika Kambhammettu

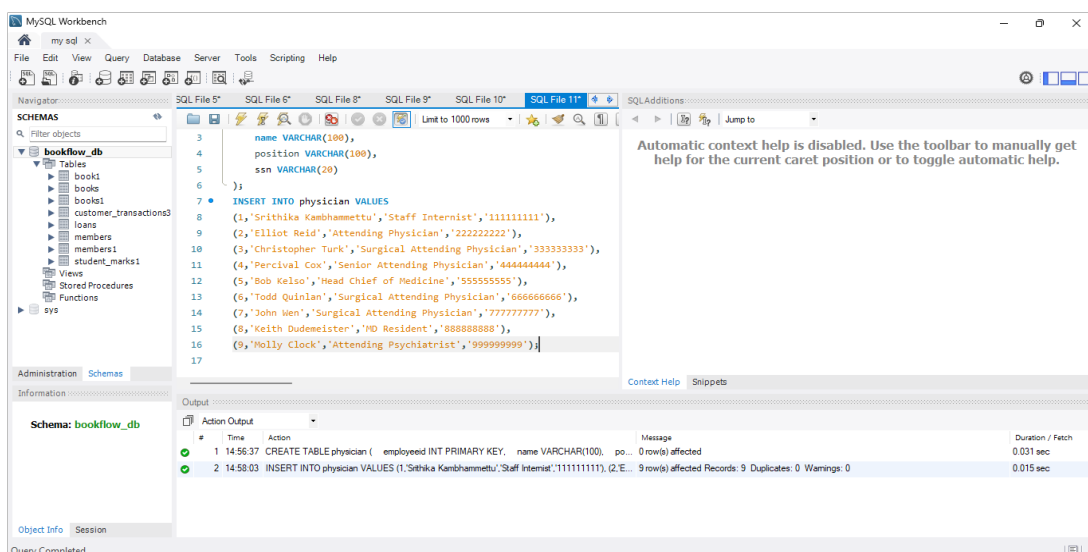
Roll no:- 2520030361

## 5 – PD Queries

### STEP 1: CREATE PHYSICIAN TABLE



### STEP 2: INSERT DATA INTO PHYSICIAN



## STEP 3: CREATE DEPARTMENT TABLE

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema with various tables and views. The main editor window shows the SQL File 11\* tab with the following SQL code:

```
CREATE TABLE department (  
  departmentid INT PRIMARY KEY,  
  name VARCHAR(100),  
  head INT  
);
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom output window shows the execution results:

| # | Time     | Action                                                                                          | Message                                                | Duration / Fetch |
|---|----------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------|
| 1 | 14:56:37 | CREATE TABLE physician ( employeeid INT PRIMARY KEY, name VARCHAR(100), po...                   | 0 row(s) affected                                      | 0.031 sec        |
| 2 | 14:58:03 | INSERT INTO physician VALUES (1,'Sithika Kamthammettu','Staff Internist','111111111'), (2,'E... | 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0 | 0.015 sec        |
| 3 | 14:58:56 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), ...                  | 0 row(s) affected                                      | 0.047 sec        |

## STEP 4: INSERT DATA INTO DEPARTMENT

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema. The main editor window shows the SQL File 11\* tab with the following SQL code:

```
CREATE TABLE department (  
  departmentid INT PRIMARY KEY,  
  name VARCHAR(100),  
  head INT  
);  
  
INSERT INTO department VALUES  
(1,'General Medicine',4),  
(2,'Surgery',7),  
(3,'Psychiatry',9);
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom output window shows the execution results:

| # | Time     | Action                                                                                          | Message                                                | Duration / Fetch |
|---|----------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------|
| 1 | 14:56:37 | CREATE TABLE physician ( employeeid INT PRIMARY KEY, name VARCHAR(100), po...                   | 0 row(s) affected                                      | 0.031 sec        |
| 2 | 14:58:03 | INSERT INTO physician VALUES (1,'Sithika Kamthammettu','Staff Internist','111111111'), (2,'E... | 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0 | 0.015 sec        |
| 3 | 14:58:56 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), ...                  | 0 row(s) affected                                      | 0.047 sec        |
| 4 | 14:59:43 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)     | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 | 0.016 sec        |

## QUESTION 1 - INNER JOIN

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow\_db

Tables

book1

books

books1

customer\_transactions3

loans

members

members1

student\_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: bookflow\_db

Object Info Session

Query Completed

SQL File 5\* SQL File 6\* SQL File 8\* SQL File 9\* SQL File 10\* SQL File 11\*

Limit to 1000 rows

SQL

28

29 • SELECT d.name AS Department,

30 p.name AS Head\_Physician

31 FROM department d

32 JOIN physician p

33 ON d.head = p.employeeid;

34

Result Grid

Filter Rows:

Export: Wrap Cell Contents

Result Grid

Form Editor

Result 1 x

Read Only Context Help Snippets

Output

Action Output

| # | Time     | Action                                                                                        | Message                                                | Duration / Fetch      |
|---|----------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 1 | 14:56:37 | CREATE TABLE physician ( employeeid INT PRIMARY KEY, name VARCHAR(100), p...                  | 0 row(s) affected                                      | 0.031 sec             |
| 2 | 14:58:03 | INSERT INTO physician VALUES (1,'Sithika Kamhammettu','Staff Internist','111111111'), (2,'... | 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0 | 0.015 sec             |
| 3 | 14:58:56 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), ...                | 0 row(s) affected                                      | 0.047 sec             |
| 4 | 14:59:43 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)   | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 | 0.016 sec             |
| 5 | 15:00:41 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN ...              | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |

## QUESTION 2 - WHERE CLAUSE

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow\_db

Tables

book1

books

books1

customer\_transactions3

loans

members

members1

student\_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: bookflow\_db

Object Info Session

Query Completed

SQL File 5\* SQL File 6\* SQL File 8\* SQL File 9\* SQL File 10\* SQL File 11\*

Limit to 1000 rows

SQL

32 JOIN physician p

33 ON d.head = p.employeeid;

34 • SELECT \*

35 FROM physician

36 WHERE position='Surgical Attending Physician';

37

38

Result Grid

Filter Rows:

Edit: Export/Import:

Result Grid

Form Editor

physician 2 x

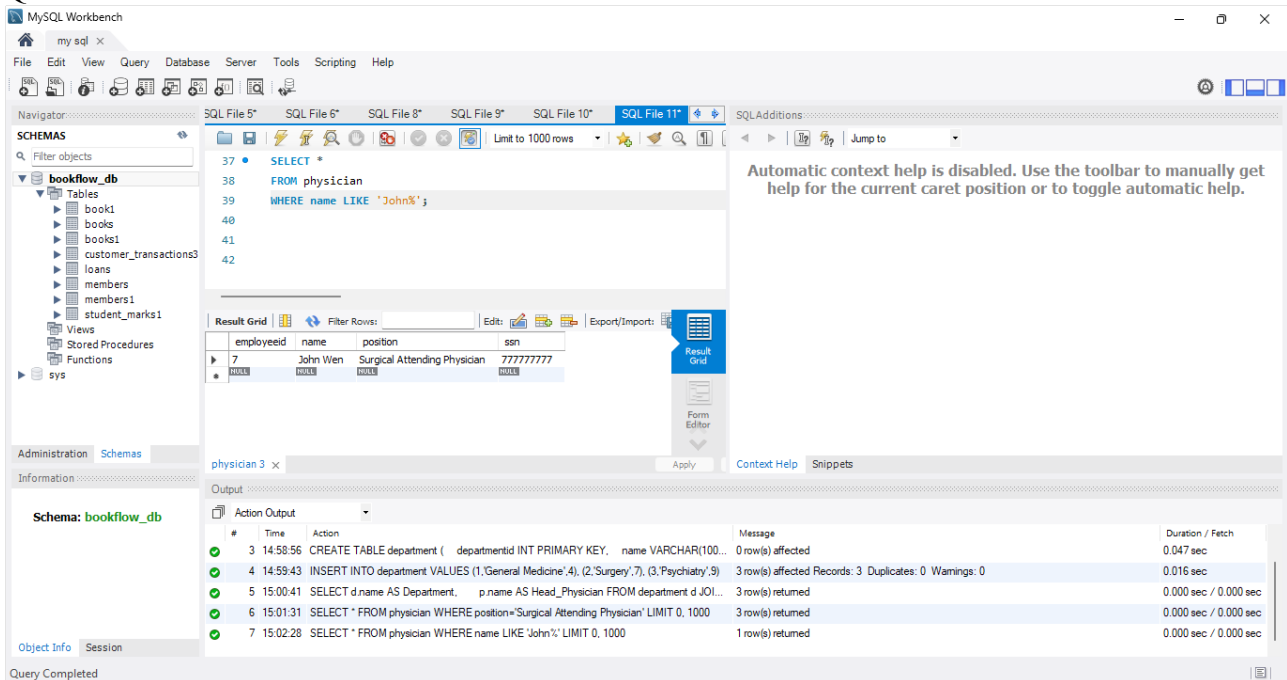
Apply Context Help Snippets

Output

Action Output

| # | Time     | Action                                                                                      | Message                                                | Duration / Fetch      |
|---|----------|---------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 2 | 14:58:03 | INSERT INTO physician VALUES (1,'Sithika Kamhammettu','Staff Internist','111111111'), (...  | 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0 | 0.015 sec             |
| 3 | 14:58:56 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), ...              | 0 row(s) affected                                      | 0.047 sec             |
| 4 | 14:59:43 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 | 0.016 sec             |
| 5 | 15:00:41 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOI...              | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |
| 6 | 15:01:31 | SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000         | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |

## QUESTION 3 - WHERE + LIKE



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema with various tables and views. The central SQL editor contains the following query:

```
37 SELECT *
38 FROM physician
39 WHERE name LIKE 'John%';
40
41
42
```

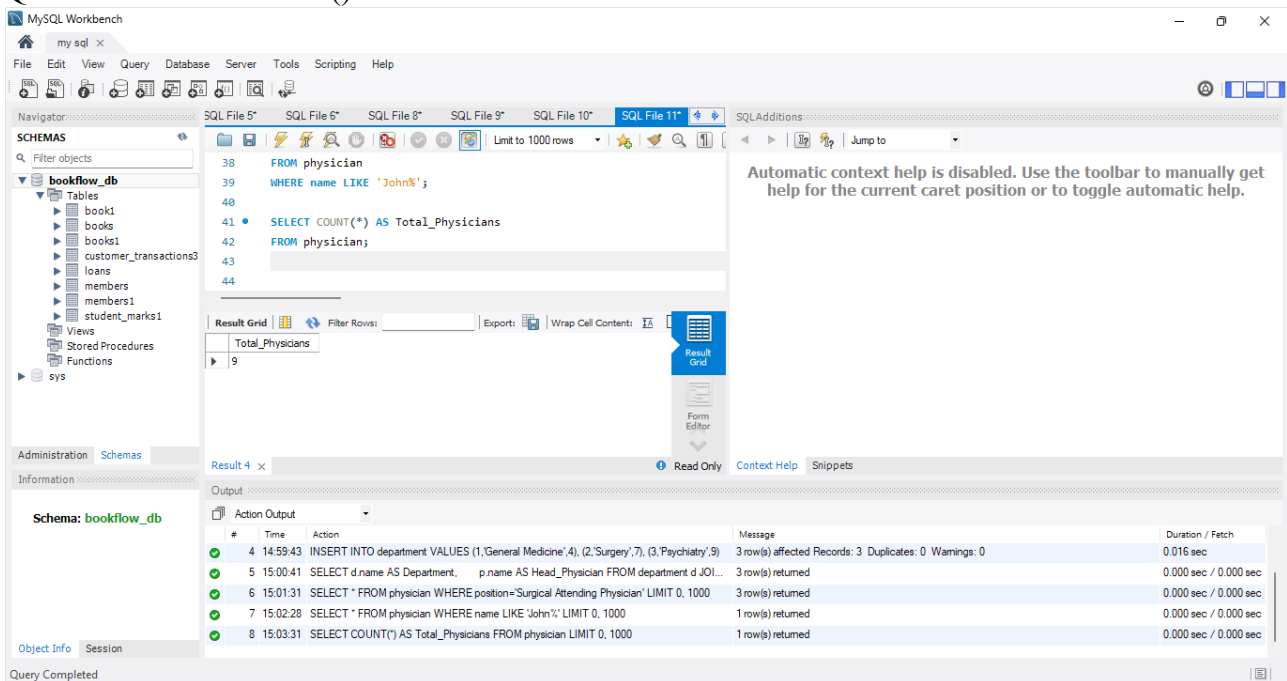
The 'Result Grid' shows the results of the query:

| employeeid | name     | position                     | ssn       |
|------------|----------|------------------------------|-----------|
| 7          | John Wen | Surgical Attending Physician | 777777777 |

The 'Output' pane at the bottom shows the 'Action Output' table:

| # | Time     | Action                                                                                      | Message                                                | Duration / Fetch      |
|---|----------|---------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 3 | 14:58:56 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100...                 | 0 row(s) affected                                      | 0.047 sec             |
| 4 | 14:59:43 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 | 0.016 sec             |
| 5 | 15:00:41 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN...             | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |
| 6 | 15:01:31 | SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000         | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |
| 7 | 15:02:28 | SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000                               | 1 row(s) returned                                      | 0.000 sec / 0.000 sec |

## QUESTION 4 – COUNT()



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema. The central SQL editor contains the following query:

```
38 FROM physician
39 WHERE name LIKE 'John%';
40
41 SELECT COUNT(*) AS Total_Physicians
42 FROM physician;
43
44
```

The 'Result Grid' shows the results of the query:

| Total_Physicians |
|------------------|
| 9                |

The 'Output' pane at the bottom shows the 'Action Output' table:

| # | Time     | Action                                                                                      | Message                                                | Duration / Fetch      |
|---|----------|---------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 4 | 14:59:43 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 | 0.016 sec             |
| 5 | 15:00:41 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN...             | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |
| 6 | 15:01:31 | SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000         | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |
| 7 | 15:02:28 | SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000                               | 1 row(s) returned                                      | 0.000 sec / 0.000 sec |
| 8 | 15:03:31 | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                            | 1 row(s) returned                                      | 0.000 sec / 0.000 sec |

## QUESTION 5 - COUNT DISTINCT

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema with various tables and views. The central SQL editor contains the following query:

```
44 SELECT COUNT(DISTINCT position)
45 FROM physician;
46
47
48
49
```

The 'Result Grid' shows a single row with the value 7.

The 'Output' pane at the bottom displays the 'Action Output' table:

| # | Time     | Action                                                                              | Message           | Duration / Fetch      |
|---|----------|-------------------------------------------------------------------------------------|-------------------|-----------------------|
| 5 | 15:00:41 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN...     | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 6 | 15:01:31 | SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000 | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 7 | 15:02:28 | SELECT * FROM physician WHERE name LIKE 'John%': LIMIT 0, 1000                      | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 8 | 15:03:31 | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                    | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 9 | 15:04:02 | SELECT COUNT(DISTINCT position) FROM physician LIMIT 0, 1000                        | 1 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 6 - GROUP BY

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema. The central SQL editor contains the following query:

```
45 FROM physician;
46 SELECT position,
47 COUNT(*) AS Total_Employees
48 FROM physician
49 GROUP BY position;
50
51
```

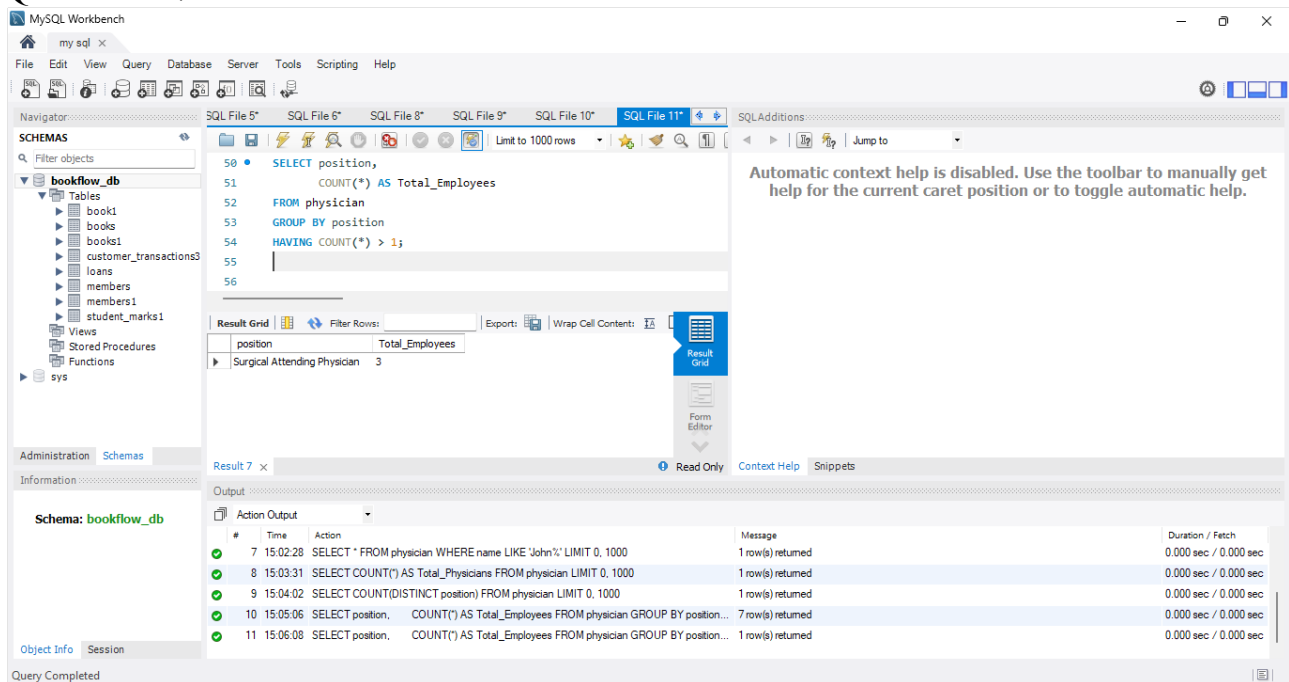
The 'Result Grid' shows the following data:

| position                     | Total_Employees |
|------------------------------|-----------------|
| Staff Internist              | 1               |
| Attending Physician          | 1               |
| Surgical Attending Physician | 3               |
| Senior Attending Physician   | 1               |
| Head Chief of Medicine       | 1               |
| MD Resident                  | 1               |
| Attending Psychiatrist       | 1               |

The 'Output' pane at the bottom displays the 'Action Output' table:

| #  | Time     | Action                                                                              | Message           | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------|-------------------|-----------------------|
| 6  | 15:01:31 | SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000 | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 7  | 15:02:28 | SELECT * FROM physician WHERE name LIKE 'John%': LIMIT 0, 1000                      | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 8  | 15:03:31 | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                    | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 9  | 15:04:02 | SELECT COUNT(DISTINCT position) FROM physician LIMIT 0, 1000                        | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 10 | 15:05:06 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position...    | 7 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 7 - GROUP BY + HAVING



MySQL Workbench interface showing a query in SQL File 11\*. The query is:

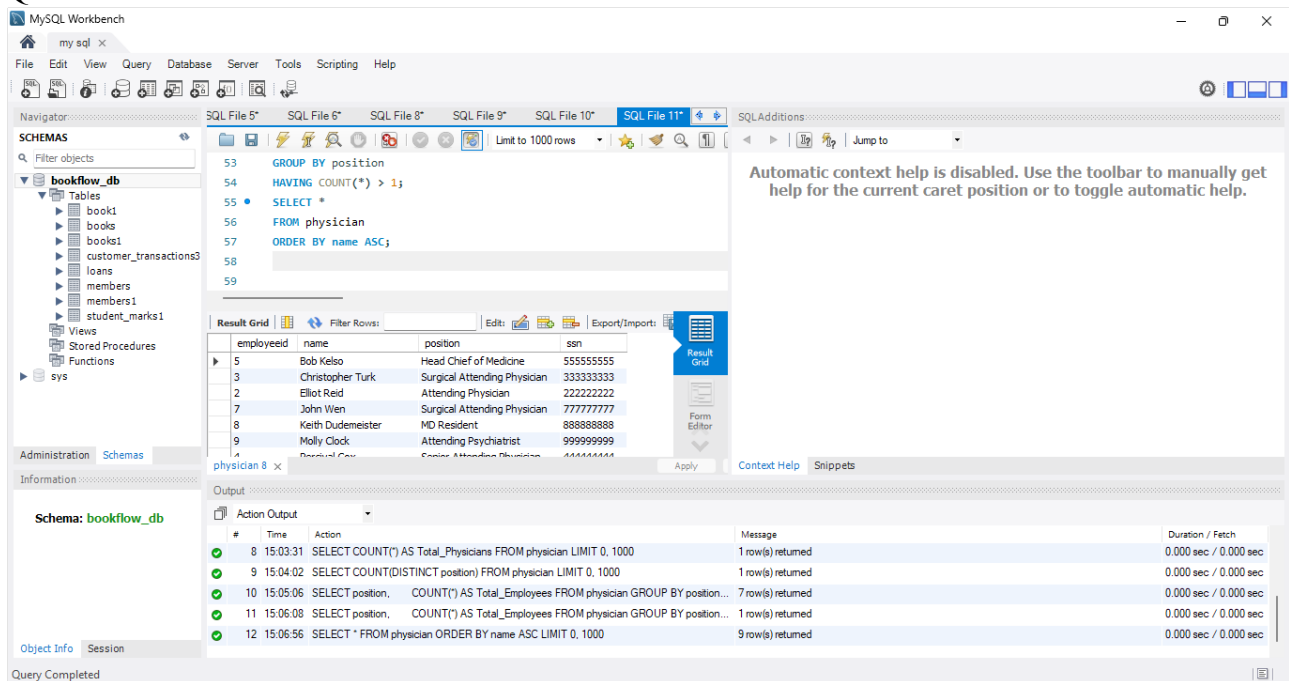
```
50 SELECT position,  
51 COUNT(*) AS Total_Employees  
52 FROM physician  
53 GROUP BY position  
54 HAVING COUNT(*) > 1;  
55  
56
```

The Result Grid shows the following data:

| position                     | Total_Employees |
|------------------------------|-----------------|
| Surgical Attending Physician | 3               |

The Output pane shows the Action Output for the query, including the message "1 row(s) returned" and the duration "0.000 sec / 0.000 sec".

## QUESTION 8 - ORDER BY



MySQL Workbench interface showing a query in SQL File 11\*. The query is:

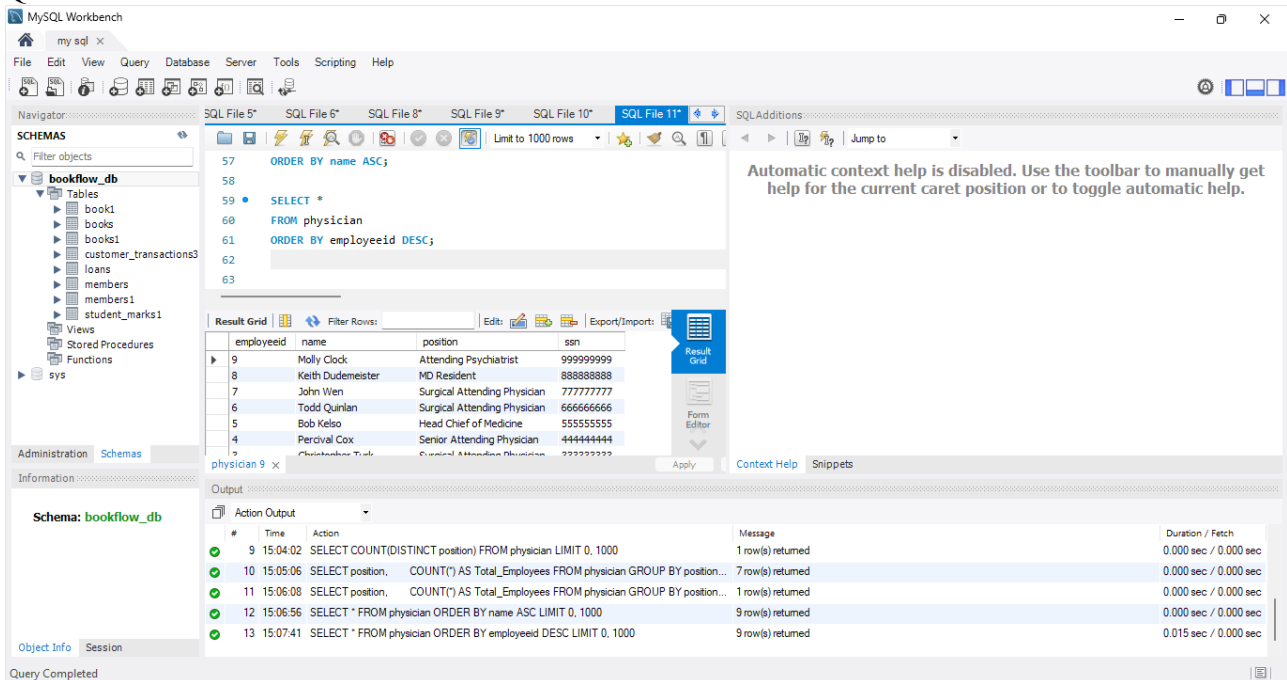
```
53 GROUP BY position  
54 HAVING COUNT(*) > 1;  
55  
56 SELECT *  
57 FROM physician  
58 ORDER BY name ASC;  
59
```

The Result Grid shows the following data:

| employeeid | name              | position                     | ssn      |
|------------|-------------------|------------------------------|----------|
| 5          | Bob Kelso         | Head Chief of Medicine       | 55555555 |
| 3          | Christopher Turk  | Surgical Attending Physician | 33333333 |
| 2          | Elliot Reid       | Attending Physician          | 22222222 |
| 7          | John Wen          | Surgical Attending Physician | 77777777 |
| 8          | Keith Dudemeister | MD Resident                  | 88888888 |
| 9          | Molly Clock       | Attending Psychiatrist       | 99999999 |
| 4          | Barbara Cor       | Senior Attending Physician   | 44444444 |

The Output pane shows the Action Output for the query, including the message "9 row(s) returned" and the duration "0.000 sec / 0.000 sec".

## QUESTION 9 - ORDER BY DESC



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema with various tables and views. The central editor window contains the following SQL query:

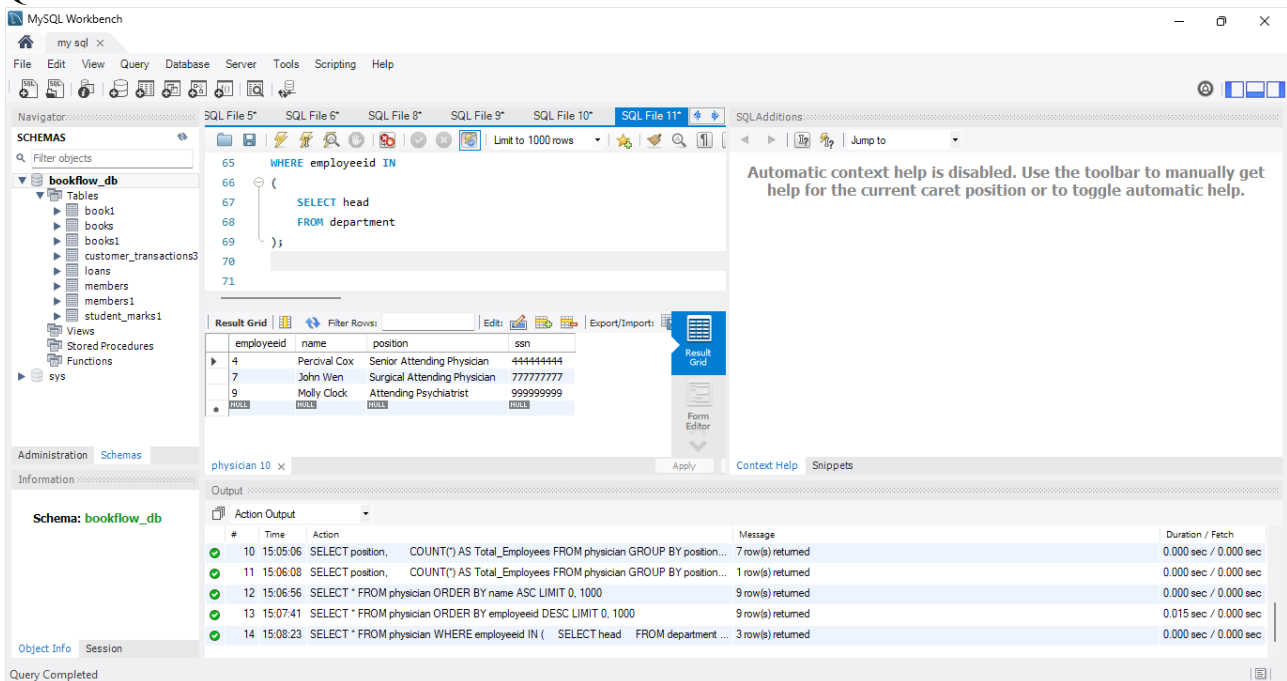
```
57 ORDER BY name ASC;
58
59 • SELECT *
60 FROM physician
61 ORDER BY employeeid DESC;
62
63
```

The 'Result Grid' shows the output of the query, displaying a list of physicians ordered by employee ID in descending order:

| employeeid | name              | position                     | ssn       |
|------------|-------------------|------------------------------|-----------|
| 9          | Molly Clock       | Attending Psychiatrist       | 999999999 |
| 8          | Keith Dudemeister | MD Resident                  | 888888888 |
| 7          | John Wen          | Surgical Attending Physician | 777777777 |
| 6          | Todd Quinlan      | Surgical Attending Physician | 666666666 |
| 5          | Bob Kelso         | Head Chief of Medicine       | 555555555 |
| 4          | Perovial Cox      | Senior Attending Physician   | 444444444 |

The 'Output' pane at the bottom shows the execution log, including the query execution time and the number of rows returned for each step.

## QUESTION 10 - WHERE + IN



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema. The central editor window contains the following SQL query:

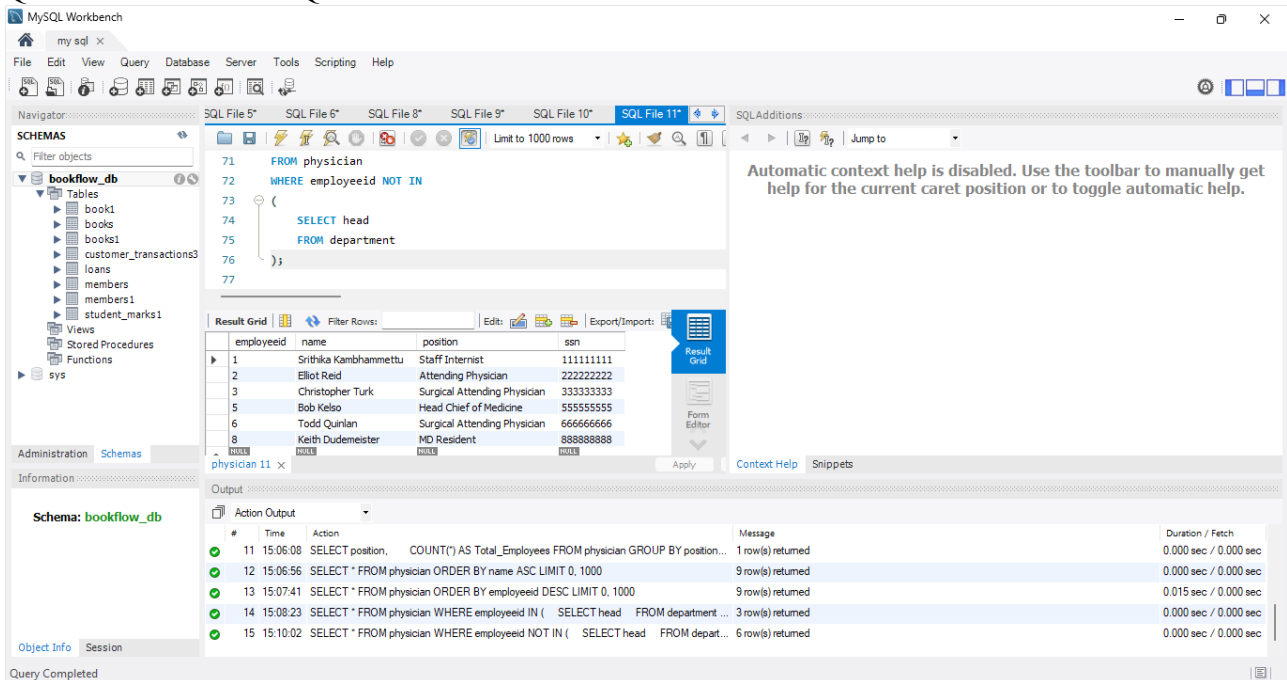
```
65 WHERE employeeid IN
66 (
67 SELECT head
68 FROM department
69 );
70
71
```

The 'Result Grid' shows the output of the query, displaying a list of physicians where the employee ID is in the result set of the subquery:

| employeeid | name         | position                     | ssn       |
|------------|--------------|------------------------------|-----------|
| 4          | Perovial Cox | Senior Attending Physician   | 444444444 |
| 7          | John Wen     | Surgical Attending Physician | 777777777 |
| 9          | Molly Clock  | Attending Psychiatrist       | 999999999 |

The 'Output' pane at the bottom shows the execution log, including the query execution time and the number of rows returned for each step.

## QUESTION 11 – SUBQUERY



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow\_db

Tables

book1

books

books1

customer\_transactions3

loans

members

members1

student\_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: bookflow\_db

Object Info Session

SQL File 5\* SQL File 6\* SQL File 8\* SQL File 9\* SQL File 10\* SQL File 11\*

Limit to 1000 rows

SQL

71 FROM physician

72 WHERE employeeid NOT IN

73

74 (

75 SELECT head

76 FROM department

77 )

Result Grid

| employeeid | name                  | position                     | ssn       |
|------------|-----------------------|------------------------------|-----------|
| 1          | Srithika Kambhammettu | Staff Internist              | 111111111 |
| 2          | Elliot Reid           | Attending Physician          | 222222222 |
| 3          | Christopher Turk      | Surgical Attending Physician | 333333333 |
| 5          | Bob Kelso             | Head Chief of Medicine       | 555555555 |
| 6          | Todd Quinlan          | Surgical Attending Physician | 666666666 |
| 8          | Keith Dudemeister     | MD Resident                  | 888888888 |

physician 11 x

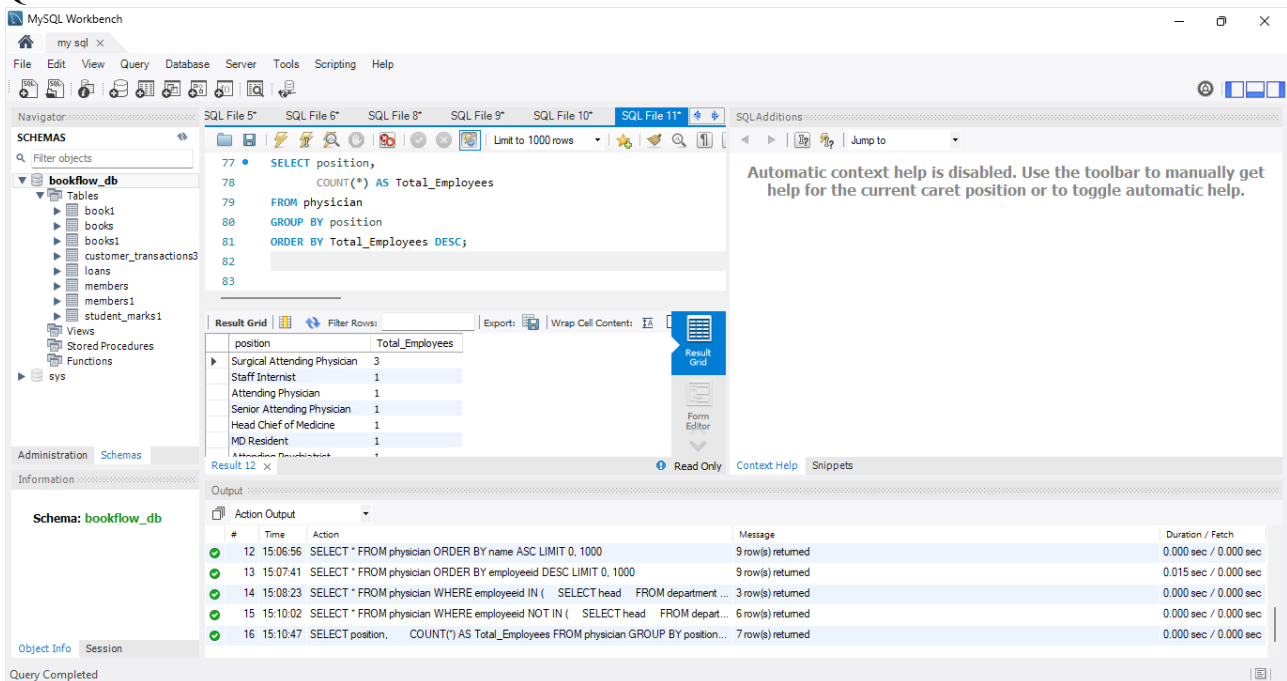
Output

Action Output

| #  | Time     | Action                                                                           | Message           | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------------|-------------------|-----------------------|
| 11 | 15:06:08 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position... | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 12 | 15:06:56 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                          | 9 row(s) returned | 0.000 sec / 0.000 sec |
| 13 | 15:07:41 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                   | 9 row(s) returned | 0.015 sec / 0.000 sec |
| 14 | 15:08:23 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ...    | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 15 | 15:10:02 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM depart...     | 6 row(s) returned | 0.000 sec / 0.000 sec |

Query Completed

## QUESTION 12 - GROUP BY + ORDER BY



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow\_db

Tables

book1

books

books1

customer\_transactions3

loans

members

members1

student\_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: bookflow\_db

Object Info Session

SQL File 5\* SQL File 6\* SQL File 8\* SQL File 9\* SQL File 10\* SQL File 11\*

Limit to 1000 rows

SQL

77 SELECT position,

78 COUNT(\*) AS Total\_Employees

79 FROM physician

80 GROUP BY position

81 ORDER BY Total\_Employees DESC;

82

83

Result Grid

| position                     | Total_Employees |
|------------------------------|-----------------|
| Surgical Attending Physician | 3               |
| Staff Internist              | 1               |
| Attending Physician          | 1               |
| Senior Attending Physician   | 1               |
| Head Chief of Medicine       | 1               |
| MD Resident                  | 1               |

Result 12 x

Output

Action Output

| #  | Time     | Action                                                                           | Message           | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------------|-------------------|-----------------------|
| 12 | 15:06:56 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                          | 9 row(s) returned | 0.000 sec / 0.000 sec |
| 13 | 15:07:41 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                   | 9 row(s) returned | 0.015 sec / 0.000 sec |
| 14 | 15:08:23 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ...    | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 15 | 15:10:02 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM depart...     | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 16 | 15:10:47 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position... | 7 row(s) returned | 0.000 sec / 0.000 sec |

Query Completed

## QUESTION 13 - WHERE + LIKE

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema with various tables and views. The central editor window contains the following SQL query:

```
81 ORDER BY Total_Employees DESC;  
82  
83 • SELECT *  
84 FROM physician  
85 WHERE position LIKE '%Attending%';  
86  
87
```

The 'Result Grid' shows the following data:

| employeeid | name             | position                     | ssn       |
|------------|------------------|------------------------------|-----------|
| 2          | Elliot Reid      | Attending Physician          | 222222222 |
| 3          | Christopher Turk | Surgical Attending Physician | 333333333 |
| 4          | Percival Cox     | Senior Attending Physician   | 444444444 |
| 6          | Todd Quinlan     | Surgical Attending Physician | 666666666 |
| 7          | John Wen         | Surgical Attending Physician | 777777777 |
| 9          | Molly Clock      | Attending Psychiatrist       | 999999999 |

The 'Output' pane at the bottom shows the execution log, including the query execution time and the number of rows returned.

## QUESTION 14 - JOIN + WHERE

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema. The central editor window contains the following SQL query:

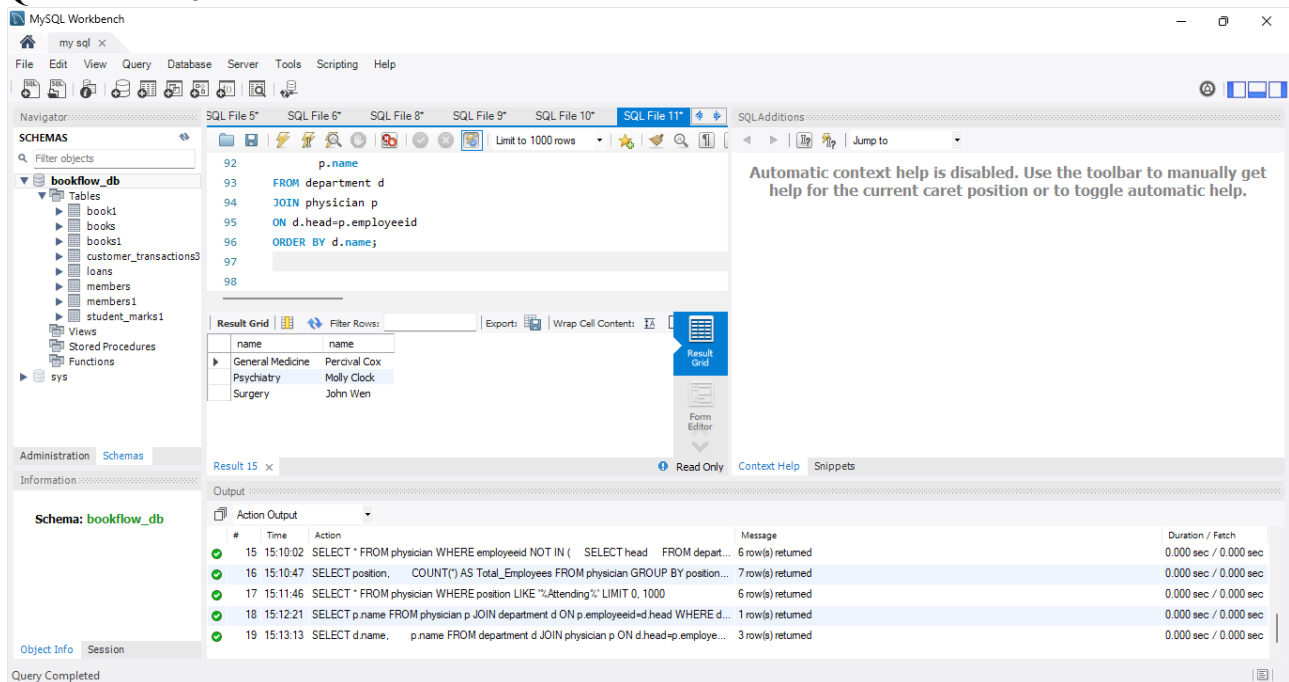
```
86 • SELECT p.name  
87 FROM physician p  
88 JOIN department d  
89 ON p.employeeid=d.head  
90 WHERE d.name='Surgery';  
91  
92
```

The 'Result Grid' shows the following data:

| name     |
|----------|
| John Wen |

The 'Output' pane at the bottom shows the execution log, including the query execution time and the number of rows returned.

## QUESTION 15 - JOIN + ORDER BY



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema with various tables and views. The central SQL editor contains a query for Question 15, which joins the 'physician' and 'department' tables and orders the results by department name. The query is as follows:

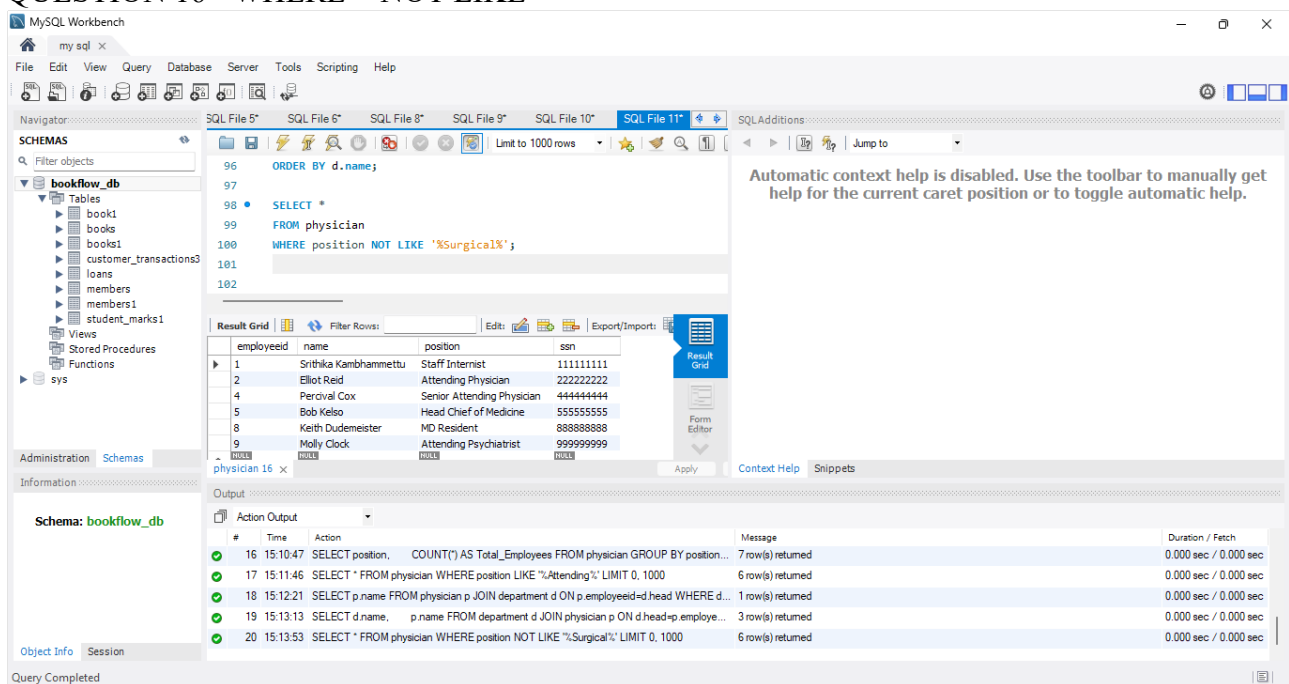
```
92 p.name
93 FROM department d
94 JOIN physician p
95 ON d.head=p.employeeid
96 ORDER BY d.name;
```

The 'Result Grid' shows the output of the query, displaying the names of physicians and their corresponding departments:

| name             | name         |
|------------------|--------------|
| General Medicine | Percival Cox |
| Psychiatry       | Molly Clock  |
| Surgery          | John Wen     |

The bottom 'Output' pane shows the execution log for the query, indicating that 3 rows were returned.

## QUESTION 16 - WHERE + NOT LIKE



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema. The central SQL editor contains a query for Question 16, which selects all physicians whose position is not 'Surgical'. The query is as follows:

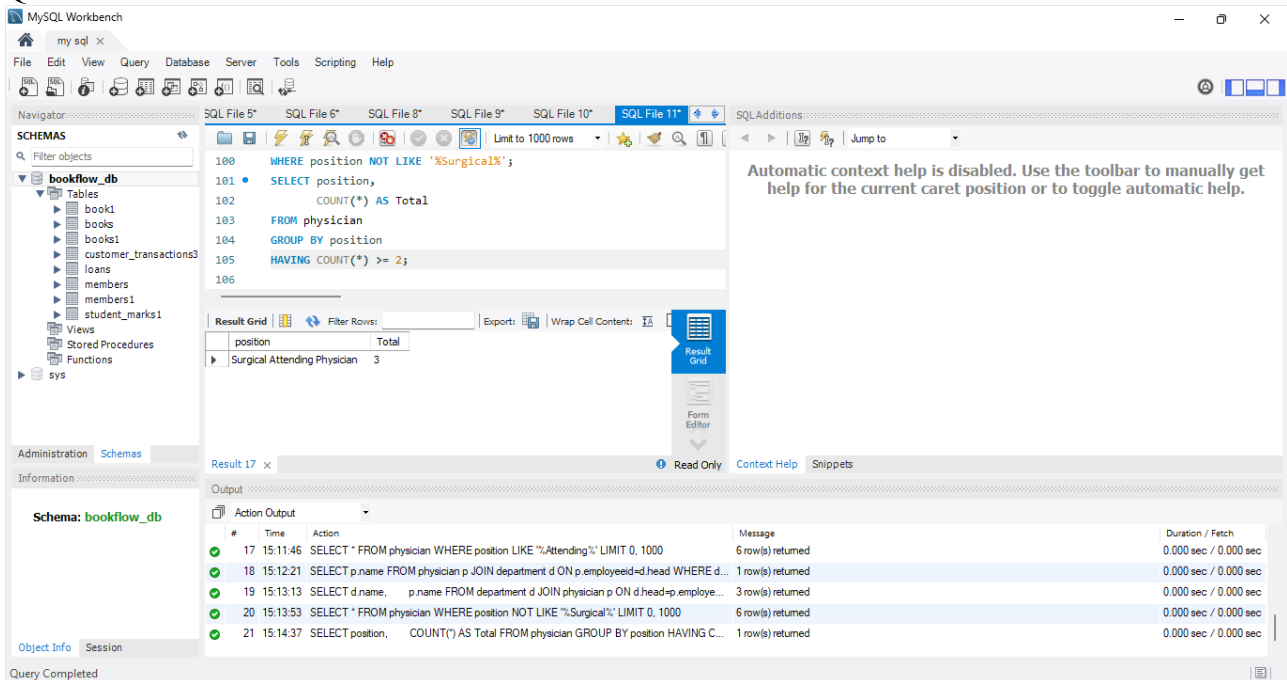
```
96 ORDER BY d.name;
97
98 SELECT *
99 FROM physician
100 WHERE position NOT LIKE '%Surgical%';
101
102
```

The 'Result Grid' shows the output of the query, displaying a list of physicians and their positions:

| employeeid | name                  | position                   | ssn       |
|------------|-----------------------|----------------------------|-----------|
| 1          | Srithika Kambhammettu | Staff Internist            | 111111111 |
| 2          | Elliott Reid          | Attending Physician        | 222222222 |
| 4          | Percival Cox          | Senior Attending Physician | 444444444 |
| 5          | Bob Kelso             | Head Chief of Medicine     | 555555555 |
| 8          | Keith Dudemeister     | MD Resident                | 888888888 |
| 9          | Molly Clock           | Attending Psychiatrist     | 999999999 |

The bottom 'Output' pane shows the execution log for the query, indicating that 6 rows were returned.

## QUESTION 17 - COUNT + HAVING



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema with various tables. The central editor window contains the following SQL query:

```
100 WHERE position NOT LIKE '%Surgical%';
101 SELECT position,
102        COUNT(*) AS Total
103 FROM physician
104 GROUP BY position
105 HAVING COUNT(*) >= 2;
106
```

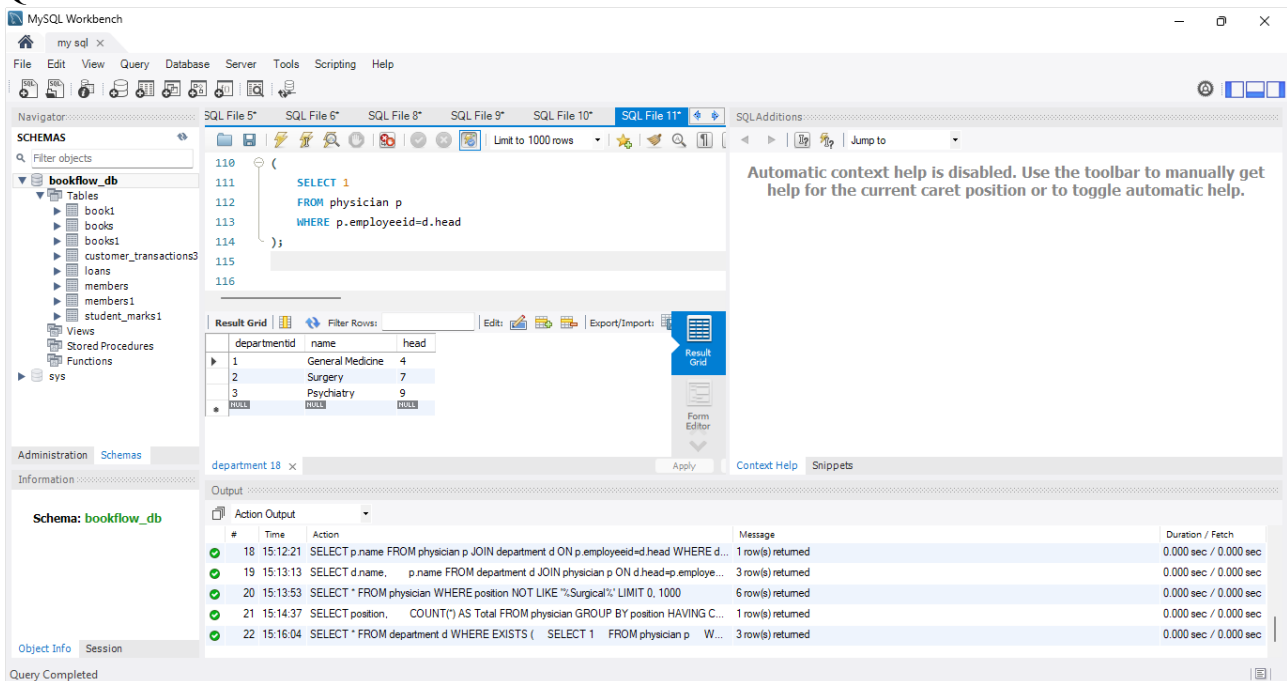
The 'Result Grid' below the query shows the following data:

| position                     | Total |
|------------------------------|-------|
| Surgical Attending Physician | 3     |

The bottom 'Output' pane shows the execution log for 'Result 17'.

| #  | Time     | Action                                                                             | Message           | Duration / Fetch      |
|----|----------|------------------------------------------------------------------------------------|-------------------|-----------------------|
| 17 | 15:11:46 | SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000            | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 18 | 15:12:21 | SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d... | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 19 | 15:13:13 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employee...   | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 20 | 15:13:53 | SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000         | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 21 | 15:14:37 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING C...    | 1 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 18 – EXISTS



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow\_db' schema. The central editor window contains the following SQL query:

```
110 (
111     SELECT 1
112     FROM physician p
113     WHERE p.employeeid=d.head
114 );
115
116
```

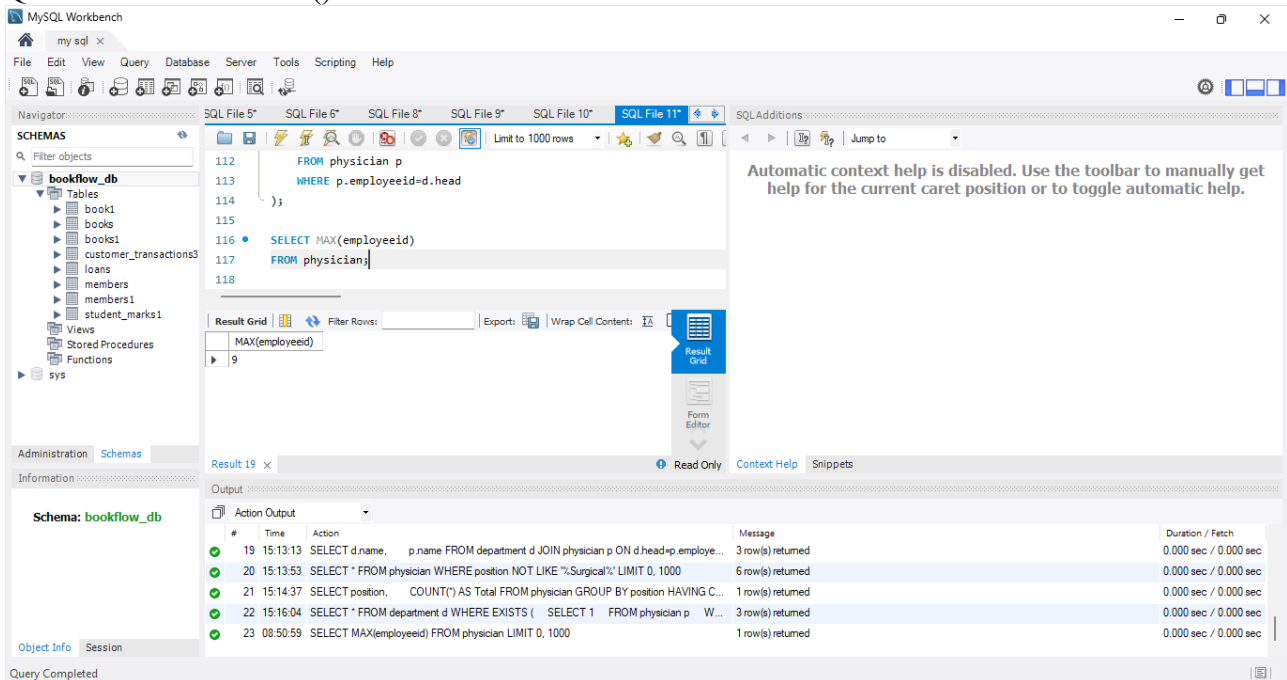
The 'Result Grid' below the query shows the following data:

| departmentid | name             | head |
|--------------|------------------|------|
| 1            | General Medicine | 4    |
| 2            | Surgery          | 7    |
| 3            | Psychiatry       | 9    |

The bottom 'Output' pane shows the execution log for 'department 18'.

| #  | Time     | Action                                                                             | Message           | Duration / Fetch      |
|----|----------|------------------------------------------------------------------------------------|-------------------|-----------------------|
| 18 | 15:12:21 | SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d... | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 19 | 15:13:13 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employee...   | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 20 | 15:13:53 | SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000         | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 21 | 15:14:37 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING C...    | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 22 | 15:16:04 | SELECT * FROM department d WHERE EXISTS ( SELECT 1 FROM physician p W...           | 3 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 19 – MAX()



The screenshot shows MySQL Workbench with the 'bookflow\_db' schema selected. The SQL editor contains the following query:

```
112 FROM physician p
113 WHERE p.employeeid=d.head
114 );
115
116 • SELECT MAX(employeeid)
117 FROM physician;
118
```

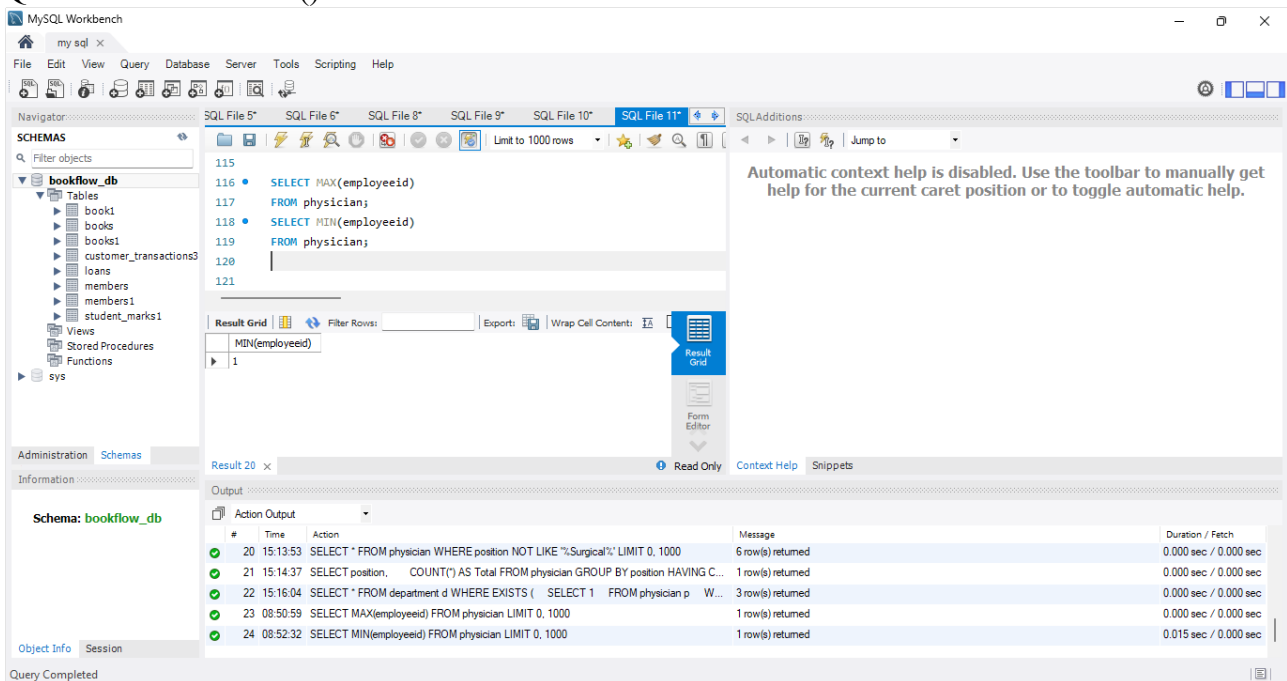
The 'Result Grid' shows the output of the query:

| MAX(employeeid) |
|-----------------|
| 9               |

The 'Output' tab shows the execution log for 'Result 19':

| #  | Time     | Action                                                                          | Message           | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------------|-------------------|-----------------------|
| 19 | 15:13:13 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 20 | 15:13:53 | SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000      | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 21 | 15:14:37 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING C... | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 22 | 15:16:04 | SELECT * FROM department d WHERE EXISTS ( SELECT 1 FROM physician p W...        | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 23 | 08:50:59 | SELECT MAX(employeeid) FROM physician LIMIT 0, 1000                             | 1 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 20 – MIN()



The screenshot shows MySQL Workbench with the 'bookflow\_db' schema selected. The SQL editor contains the following query:

```
115
116 • SELECT MAX(employeeid)
117 FROM physician;
118 • SELECT MIN(employeeid)
119 FROM physician;
120
121
```

The 'Result Grid' shows the output of the query:

| MIN(employeeid) |
|-----------------|
| 1               |

The 'Output' tab shows the execution log for 'Result 20':

| #  | Time     | Action                                                                          | Message           | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------------|-------------------|-----------------------|
| 20 | 15:13:53 | SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000      | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 21 | 15:14:37 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING C... | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 22 | 15:16:04 | SELECT * FROM department d WHERE EXISTS ( SELECT 1 FROM physician p W...        | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 23 | 08:50:59 | SELECT MAX(employeeid) FROM physician LIMIT 0, 1000                             | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 24 | 08:52:32 | SELECT MIN(employeeid) FROM physician LIMIT 0, 1000                             | 1 row(s) returned | 0.015 sec / 0.000 sec |